

杭州电子科技大学学生考试卷 (B) 卷

考试课程	通信电路与系统	考试日期	2022 年 6 月 22 日	成绩	
课程号	A040095a	教师号		任课教师姓名	陈道、林弥、李航、 刘国华、刘艳、张伟、 孙朋飞、苏江涛、关 阳阳
考生姓名		学号 (8 位)		年级	

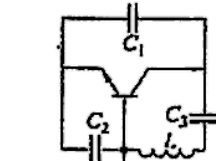
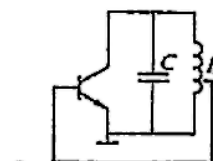
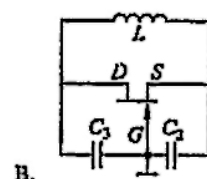
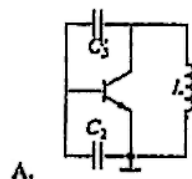
题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题	第七大题
得分							

I. Single Choice (3' each)

1. A serial LC circuit resonates at the frequency of f_0 . When the actual frequency f is larger than f_0 ($f > f_0$), the circuit shows (A).

- A. Inductive B. Capacitive C. Resistive D. Short

2. By using the phase equilibrium condition, determine which of the oscillation circuits shown below is possible. (D)



3. Which of the conduction angle (CCA) 2θ is classified as Class C power amplifier? (D)

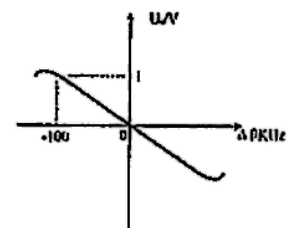
- A. $2\theta=360^\circ$
C. $2\theta=180^\circ$

- B. $180^\circ < 2\theta < 360^\circ$
D. $2\theta < 180^\circ$

4. What is not the reason why Quartz crystal has high frequency stability (B)

- A. It has low temperature coefficient
B. It can be seen as capacitor
C. It has high Q-factor.
D. It has low access factor.

5. The frequency discrimination characteristics curve has been shown below. Known that the output voltage of the frequency discriminator is $u_o(t) = 0.5 \cos 4\pi \times 10^3 t$ (V), the frequency discriminating trans-conductance g_u and maximum frequency shift Δf_m of input signal are (A)

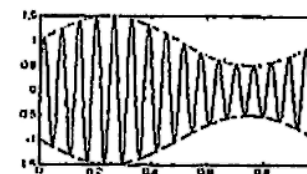


- A. $-1/100$ (V/KHz), 50 KHz B. $-1/100$ (V/KHz), 100 KHz
C. $-1/200$ (V/KHz), 50 KHz D. $-1/200$ (V/KHz), 100 KHz

6. Which of the following circuits is not part of AM receiving system? (D)

- A. Local Oscillator B. Detector
C. High Frequency Amplifier D. Modulator

7. For the standard AM signal shown below, the modulation index m_a is (C)



- A. 1
C. 0.5

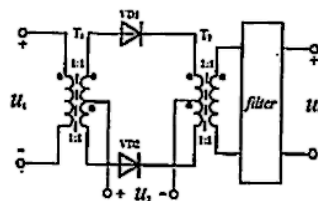
- B. 0.2
D. 2

8. A broadcasting radio signal at 550 kHz is received by a receiver with a 465 kHz IF(intermediate frequency). Another signal at 1480 kHz is also received at the same time.

Why does the 1480 kHz signal appear? (A)

- A. Mirror image interference
B. IF interference
C. Higher order side-channel interference
D. Random Noise

9. The diode multiplier circuit is shown below. Assume the I/V curve characteristics of diode VD1 and VD2 are exactly the same. $u_1 = u_{1m} \cos \omega_1 t$, $u_2 = u_{2m} \cos \omega_2 t$, $u_{2m} \gg u_{1m}$, the intermediate frequency $\omega_1 - \omega_2 - \omega_c$. What type of filter should be used at u_o terminal? (B)



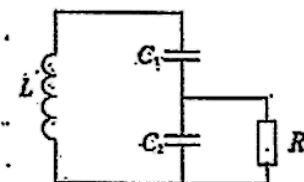
- A. Low pass filter
B. Band pass filter
C. Band stop filter
D. High pass filter

10. For standard FM wave, which of the following statement is correct? (B)

- A. The carrier amplitude will change with the modulated signal
B. The carrier frequency will change with the modulated signal
C. The carrier frequency and amplitude both change with the modulated signal
D. The carrier frequency and amplitude both remain constant

II. Calculation

1. (13') The parallel LC resonant circuit is shown below with $L=20\mu H$, $C_1=C_2=10pF$, $Q_0=100$, $R_L=20k\Omega$, please calculate the resonant frequency f_0 , the resonant resistance R_o , quality factor load Q_L , bandwidth $B_{0.7}$.



$$C_z = \frac{C_1 \cdot C_2}{C_1 + C_2} = 5pF \quad (2')$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 15.9MHz \quad (2')$$

$$R_o = Q_0 \omega_0 L = 200K\Omega \quad (2')$$

$$n = \frac{C_1}{C_1 + C_2} = \frac{1}{2} \quad (1')$$

$$R'_L = \frac{1}{n^2} R_L = 4R_L = 80K\Omega \quad (2')$$

$$Q_L = \frac{R_o}{\omega_0 L} = \frac{R_o / R'_L}{\omega_0 L} \approx 28.6 \quad (2')$$

$$B_{0.7} = \frac{f_0}{Q_L} = 556kHz \quad (2')$$

2. (14') A resonant power amplifier with operating at critical point state. Assuming $E_c = 24$ V, current conduction angle $\theta = 60^\circ$, $\alpha_0(60^\circ) = 0.218$, $\alpha_1(60^\circ) = 0.391$, collector current pulse maximum value $I_{cm} = 2.2$ A, collector voltage utilization factor $\xi = 0.9$.

(1) Please calculate output power P_{out} , power of power supply P_{DC} , collector efficiency η_c and collector equivalent load resistor R_c .

(2) If the equivalent load resistor R_c changing suddenly to short. How to change of output power P_{out} , power supply P_{DC} . Please analyze the reasons of changing and potential damage.

$$U_{c1m} = \xi \times E_c = 0.9 \times 24 = 21.6V \quad (1')$$

$$I_{c1m} = \alpha_1(\theta) \times I_{cm} = 0.391 \times 2.2 \approx 0.86A \quad (1')$$

$$P_{out} = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 21.6 \times 0.86 \approx 9.29W \quad (2')$$

$$I_{c0} = \alpha_0(\theta) \times I_{cm} = 0.218 \times 2.2 = 0.48A \quad (1')$$

$$P_{DC} = E_c \times I_{c0} = 24 \times 0.48 \approx 11.52W \quad (1')$$

$$\eta_c = \frac{P_{out}}{P_{DC}} = \frac{9.29}{11.52} \approx 80.6\% \quad (2')$$

$$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{21.6}{0.86} \approx 25.1\Omega \quad (1')$$

其它条件不变时, 集电极等效负载电阻 R_c 短路, 功放工作到欠压状态, 输出功率也会降低至 0; (3')

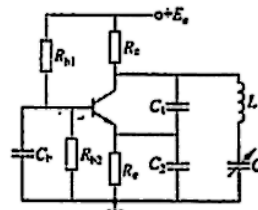
原因是输出电压基波分量会迅速减小到 0, 虽然集电极电流基波分量会缓慢增加, 但二者乘积仍然为 0, 故输出功率为 0。所有的电源功率都用来发热, 严重时烧毁功放管。(2')

3. (11') The AC equivalent circuit of Clapp oscillator is shown below. Please answer the following questions:

(1) Assuming the value of $L = 100 \mu H$, $C_1 = C_2 = 2000$ pF, $C = 100 \sim 120$ pF, calculate the resonant frequency f_0 .

(2) Calculate the feedback coefficient.

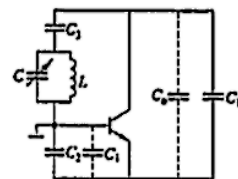
(3) To raise the frequency stability, based on Clapp oscillator, the Seiler oscillator can be obtained. Please draw the AC equivalent circuit of the Seiler oscillator circuit. And write the advantages of Seiler oscillator relative to Clapp oscillator.



$$(1) f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{LC}} \times \frac{1}{2\pi \times \sqrt{100 \times 10^{-6} \times (100 \sim 120) \times 10^{-12}}} = (1.45 \sim 1.59) \times 10^6 \text{ Hz} \quad (3')$$

$$(2) F = U_f/U_o = C_1/(C_2 + C_1) = 1/2 \quad (2')$$

(3)



(可不画 C_1, C_2)

(3')

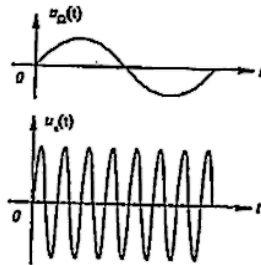
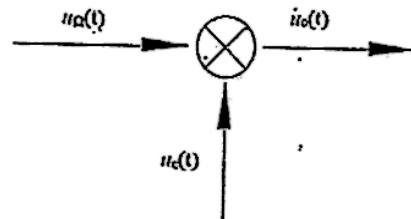
(1) Output amplitude is relatively stable when the oscillation frequency is tuned;

(2) Tuning range is relatively wide, the coverage coefficient of frequency is usually high, generally about 1.6 to 1.8. In practice, it has been widely used in broadband system. (3')

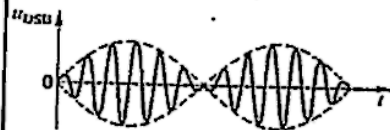
4. (11') The realization of $u_o(t)$ is shown below. $u_D(t) = U_{DM} \cos(\Omega t)$ and $u_c(t) = U_{cm} \cos(\omega_c t)$ are modulation signal and high-frequency carrier, respectively. Please answer the following questions:

(1) If $u_i(t) = \cos 100\pi t \times \cos 1000\pi t$ (v), please draw the waves of $u_D(t)$, $u_C(t)$ and $u_o(t)$, respectively. Please draw the frequency spectrum of $u_o(t)$.

(2) Calculate the total power consumed on the unit resistance.

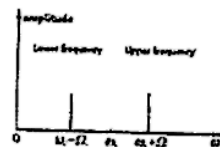


具体图形可自行判断)



(各 1', 共 2')

(3')



(注意横纵坐标)

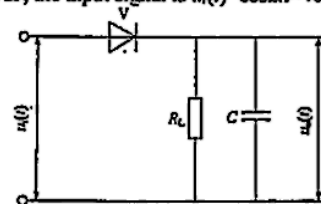
(3)

(2) $u_i(t) = \cos 100\pi t \times \cos 1000\pi t = 1/2 \times (\cos 900\pi t + \cos 1100\pi t)$

$$P = \frac{1}{2} \times \left(\frac{1}{2}\right)^2 + \frac{1}{2} \times \left(\frac{1}{2}\right)^2 = \frac{1}{4} \text{ W}$$

(3)

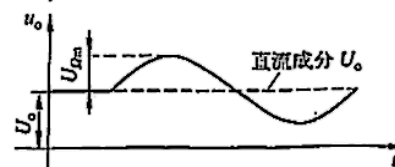
5. (12') The circuit of large signal peak envelope detector is shown as below, assuming $C = 0.05 \mu\text{F}$, the input signal is $u_i(t) = \cos 2\pi \times 465 \times 10^3 t + 0.5 \cos 2\pi \times 10^3 t \cos 2\pi \times 465 \times 10^3 t$ (v).



- (1) Please briefly describe the working principle of the circuit.
- (2) Calculate the modulation factor m_a . After filtering, draw the output signal waveform.
- (3) How to choose the load resistance R_L to prevent diagonal distortion?
- (4) How to choose the load resistance R_L to acquire high detection efficiency?

(1) 二极管工作在开关状态, 正向导通电阻很小, 当 $u_i > u_o$ 时, 电容 C 充电很快。电阻 R_L 很大, 当 $u_i < u_o$ 时, 电容放电较慢。利用快充慢放的原理使输出信号跟随输入信号包络的变化, 并最终得到调制信号, 完成检波。 (3' 画图解释亦可)

(2) $u_i(t) = \cos 2\pi \times 465 \times 10^3 t (1 + 0.5 \cos 2\pi \times 10^3 t)$ $m_a = 0.5$ (2')



(2')

(3)

$$\Omega = 2\pi \times 10^3 \text{ rad/s}$$

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} = \frac{0.75}{0.5} \times \frac{1}{2\pi \times 10^3 \times 0.05 \times 10^{-4}} \Omega = 4.77 \text{ k}\Omega \quad (3')$$

$$(4) \omega_C = 2\pi \times 465 \times 10^3 \text{ rad/s}$$

$$R_L \gg \frac{1}{\omega_C C} = \frac{1}{2\pi \times 465 \times 10^3 \times 0.05 \times 10^{-4}} \Omega = 21.5 \Omega \quad (2')$$

6. (9') Known that the modulation frequency $u_\Omega(t) = 2.5 \cos 4\pi \times 10^3 t$ (v), the carrier signal $u_c(t) = 10 \cos 2\pi \times 10^6 t$ (v), and maximum frequency shift $\Delta f_{\max} = 100 \text{ kHz}$, please:

- (1) Calculate the maximum phase shift m_f .
- (2) Write down the mathematical expression of the output FM signal $u_{FM}(t)$.
- (3) Calculate the effective bandwidth B_{FM} .

$$(1) m_f = \Delta f_{\max} / f = 100 \times 10^3 \text{ Hz} / 2000 \text{ Hz} = 50 \quad (3')$$

$$(2) u_{FM}(t) = U_m \cos(\omega_c t + m_f \sin \Omega t) = 10 \cos(2\pi \times 10^6 t + 50 \sin 4000\pi t) \quad (3')$$

$$(3) B_{FM} = 2(m_f + 1)f = 2 \times (50 + 1) \times 2000 \text{ Hz} = 204 \text{ kHz} \quad (3')$$